

RSUF: IMPROVING FLOW-BASED SPEECH GENERATION WITH RECONSTRUCTED-SPEECH UNDERSTANDING FEEDBACK

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ABSTRACT

Unified speech models commonly couple understanding and generation through parameter sharing and mixed-task training. When transcription supervision is applied only to real speech, it influences generation through shared parameters without directly assessing generated content. We propose Reconstructed-Speech Understanding Feedback (RSUF), which provides direct transcript supervision for flow-based speech generation. Starting from a flow-training state, RSUF differentially reconstructs speech latents and feeds them to a frozen understanding branch for transcript prediction. Transcript-loss gradients propagate through the reconstruction trajectory to the generator, encouraging the reconstructed speech to preserve the target text. RSUF requires no waveform decoding in the feedback branch and adds no computation at inference. Controlled bilingual experiments show that retaining shared understanding supervision reduces Chinese character error rate and English word error rate by up to 12.7% and 10.4%, respectively, relative to generation-only continuation. In a separate matched continuation, RSUF reduces these errors by 6.4% and 8.0% relative to shared understanding supervision, while improving predicted naturalness and speaker similarity. These results support using learned speech understanding to directly supervise reconstructed content beyond parameter sharing and mixed-task training. Audio samples are available on the project webpage.

Index Terms— speech generation, understanding supervision, flow matching, unified understanding and generation modeling, multi-task learning

1. INTRODUCTION

Unified speech models learn recognition and synthesis within a common architecture. Joint training offers a route for transferring linguistic knowledge to generation, but sharing parameters does not itself supervise generated content. The challenge is to turn learned speech understanding into a training signal that directly assesses whether reconstructed speech conveys the intended text. For example, fluent speech may omit or substitute target words. Transcript feedback can expose these errors and complement the acoustic generation objective.

Existing approaches connect recognition and synthesis through shared representations across several generation architectures. SpeechT5 [1] learns shared representations through encoder-decoder pretraining, while ViOLA [2] unifies recognition, synthesis, and translation through autoregressive codec language modeling. UniVoice [3] combines autoregressive recognition with flow-based synthesis in a

shared model. UniAudio 2.0 and UALM extend joint learning across audio understanding and generation tasks [4, 5]. These developments establish a foundation for joint training, while transcription objectives evaluated on real speech supervise the audio-to-text mapping. Evaluating reconstructed speech with the understanding branch provides a complementary way to transfer this learned mapping to generation.

Prior work has explored learning from feedback on synthesized speech. The machine speech chain [6] reconstructs text from synthesized speech, and StyleTTS 2 [7] uses pretrained speech-model representations for adversarial supervision. Cont-SPT [8] applies ASR supervision to reconstructed audio during tokenizer pretraining. Flow matching [9], used in F5-TTS [10] and MMAudio [11], trains a model to move intermediate noisy representations toward clean audio. Flow matching supervises a velocity field, while transcript feedback evaluates the linguistic content of its integrated endpoint. Connecting these objectives requires gradients through the reconstruction path. Building on prior work on feedback from synthesized speech, we study transcript supervision within flow-based generator training.

We propose Reconstructed-Speech Understanding Feedback (RSUF) for this setting. RSUF starts from the flow-training state already sampled for flow matching and uses a frozen latent-space understanding branch to supervise a differentially reconstructed endpoint. Reusing this state connects transcript feedback to flow matching on the same example. Freezing the reader keeps the content criterion fixed as the generator changes. Transcript-loss gradients pass through the reconstruction trajectory to the generator, without waveform decoding or additional inference computation. We evaluate RSUF in a unified multimodal diffusion transformer (MMDiT) for speech understanding and generation. Two matched continuation studies separate the benefit of retaining shared understanding supervision from the additional benefit of evaluating reconstructed speech. An ablation compares frozen and jointly updated readers to test the value of a fixed understanding mapping. Separate waveform recognizers assess content accuracy, testing whether the gains extend beyond agreement with the training reader. Together, these comparisons link the supervision design to content accuracy, predicted naturalness, and speaker similarity in both languages.

2. RECONSTRUCTED-SPEECH UNDERSTANDING FEEDBACK

RSUF feeds reconstructed speech to a frozen understanding branch and returns transcript feedback to the generator (Fig. 1).

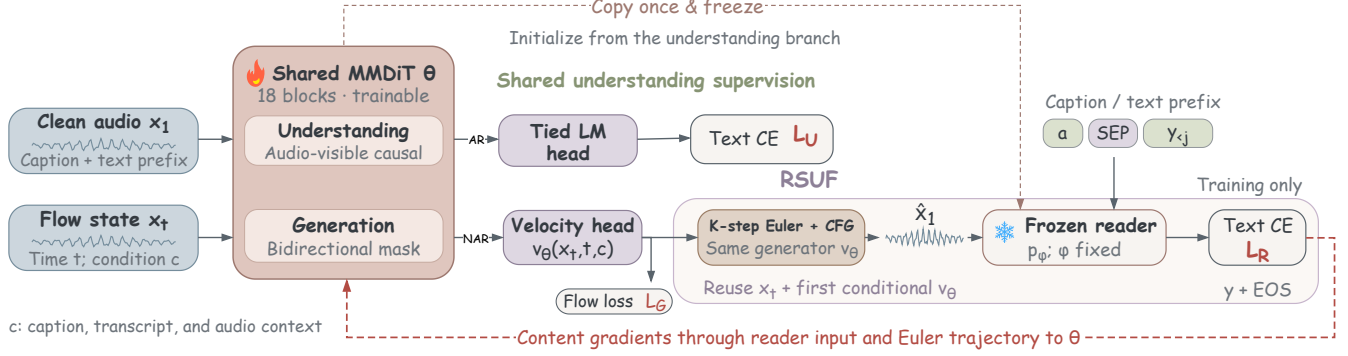


Fig. 1. RSUF within joint understanding and generation training. Shared understanding supervision predicts text from real audio. RSUF reconstructs speech latents from the flow-training state using K differentiable CFG-guided Euler steps. A frozen understanding branch receives these latents and predicts their transcript; its loss backpropagates through reconstruction to the generator. The feedback branch runs only during training. a/y : caption/transcript; AR/NAR: autoregressive/non-autoregressive. Waveforms represent audio latents.

2.1. Problem Formulation and Backbone

Let x_1 , y , and a denote clean audio latents, transcript, and caption. Generation is conditioned on $c = [a, \text{SEP}, y]$ and the available audio context. Our MMDiT has 1.69B parameters, 18 joint blocks, a hidden width of 1024, and 64-dimensional audio latents.

The two tasks share the backbone, embeddings, and projections; the text output head is tied to the input embeddings. Following UniVoice [3], generation uses bidirectional audio-text attention. For understanding, audio tokens attend only to audio, while text tokens attend to audio and preceding text, preventing future-text leakage.

2.2. Joint Understanding and Generation Training

For Gaussian noise ϵ and $t \sim \mathcal{U}(0, 1)$, flow matching uses $x_t = tx_1 + (1 - t)\epsilon$. The generation loss $\mathcal{L}_G$ is the mean squared error between $v_\theta(x_t, t, c)$ and $x_1 - \epsilon$, averaged over valid predicted frames and latent dimensions. Known audio-context frames remain fixed to their clean latents and are excluded from the prediction loss.

The same backbone learns to predict text from clean audio at $t = 1$. The understanding loss $\mathcal{L}_U$ is teacher-forced cross-entropy, with caption, separator, transcript, and EOS tokens as targets. We combine the objectives as $\mathcal{L}_{\text{joint}}^{(r)} = (r\mathcal{L}_G + \mathcal{L}_U)/(r + 1)$. Here, r controls the relative loss coefficients. Asymmetric weights prioritize generation while retaining understanding supervision. Under this normalization, moving from 10:1 to 20:1 mainly reduces the understanding coefficient and only modestly increases the generation coefficient. Reconstruction experiments start from a separate joint checkpoint, as specified in Sec. 3.1.

2.3. Differentiable Latent Reconstruction

RSUF reconstructs speech latents at $t = 1$ from the flow-training state x_t . Each reconstruction step combines conditional and unconditional velocity predictions using classifier-free guidance (CFG):

$$v_\theta^{\text{cfg}}(z, t, c) = (1 + g_{\text{rec}})v_\theta(z, t, c) - g_{\text{rec}}v_\theta(M_\emptyset z, t, c_\emptyset), \quad (1)$$

where g_{rec} is the reconstruction guidance strength. The masking operator $M_\emptyset$ zeros only known audio-context positions in z , preserving the region to be generated. For $c_\emptyset$, all text embeddings, including

caption and transcript, and any separate reference-audio latents are zeroed; lengths and masks remain unchanged.

With $z_0 = x_t$ and $h = (1 - t)/K$, we obtain $\hat{x}_1 = z_K$ through K differentiable Euler steps along the flow trajectory:

$$z_{k+1} = \Pi_c \left[z_k + h v_\theta^{\text{cfg}}(z_k, t + kh, c) \right], \quad (2)$$

for $k = 0, \dots, K - 1$. Here, Π_c restores clean known audio context and clears padding at every step before the next update.

The first Euler step reuses the conditional velocity prediction from the flow loss and adds an unconditional prediction. All subsequent steps use the same CFG rule. Both velocity branches remain differentiable throughout the trajectory.

2.4. Understanding Reconstructed Speech

A frozen copy of the understanding branch serves as the reader p_ϕ . The reader and generator are initialized from the same joint checkpoint. At $t = 1$, the reader takes $\hat{x}_1$, which combines fixed audio context with the reconstructed region. It predicts y , the transcript of the complete utterance, including both regions:

$$\mathcal{L}_R = -\frac{1}{N_y} \sum_{j \in \mathcal{T}_y} \log p_\phi(y_j | \hat{x}_1, a, y_{<j}), \quad (3)$$

where $\mathcal{T}_y$ contains transcript and EOS positions and N_y counts valid target tokens. Caption and separator tokens provide context. This loss measures how well the reader recovers the target transcript from reconstructed speech. Its gradients pass through the reader's audio input and the reconstruction trajectory to update the generator, while the audio-to-text mapping remains fixed. Samples with dropped text conditioning contribute to flow matching but are excluded from $\mathcal{L}_R$.

RSUF training optimizes

$$\mathcal{L}_{\text{recon}} = \lambda_G \mathcal{L}_G + \lambda_U \mathcal{L}_U + \lambda_R \mathcal{L}_R, \quad (4)$$

where λ_G , λ_U , and λ_R are loss weights. RSUF-off removes only $\lambda_R \mathcal{L}_R$, keeping the other coefficients fixed; the unfrozen-reader ablation also updates ϕ . The reconstruction branch operates entirely on latents and is removed at inference.

Table 1. Re-evaluated systems and converged RSUF on Seed-TTS-eval, using the same metric implementations and the generation settings in Secs. 3.2–3.3. MOS: 20 listeners $\times$ 8 utterances per system/language. Bold: best values within this table, including ties.

Model	Params (B)	Chinese				English			
		CER (%) $\downarrow$	SIM $\uparrow$	UTMOS $\uparrow$	MOS $\uparrow$	WER (%) $\downarrow$	SIM $\uparrow$	UTMOS $\uparrow$	MOS $\uparrow$
Qwen2.5-Omni-7B [12]	10.73	2.74	0.75	3.53	4.50	5.60	0.63	4.19	4.06
MiMo-Audio-7B-Instruct [13]	8.02	3.10	0.58	2.66	3.06	4.03	0.44	3.54	3.13
Step-Audio 2-mini [14]	8.32	1.41	0.64	3.51	3.88	1.89	0.47	4.31	3.75
UniAudio 2.0 [4]	4.86	5.63	0.48	3.09	3.00	4.76	0.34	3.84	3.31
Ming-Lite-Omni [15]	0.69 (Talker only)	2.05	0.61	3.73	3.19	5.62	0.45	4.24	3.75
Ming-UniAudio-16B-A3B [16]	18.27	1.06	0.70	2.77	3.56	1.82	0.58	3.67	3.44
RSUF (converged)	1.69	1.69	0.77	3.35	3.75	3.02	0.68	3.94	3.88

Table 2. Evaluating shared understanding supervision with 200k-update continuations from Joint 10:1 300k. All models have 1.69B parameters. Bold marks the best value for each metric and language within this table, including ties.

Training setting	Chinese			English		
	CER (%) $\downarrow$	SIM $\uparrow$	UTMOS $\uparrow$	WER (%) $\downarrow$	SIM $\uparrow$	UTMOS $\uparrow$
Generation-only continuation	2.52	0.73	3.12	4.23	0.67	3.76
Joint 10:1	2.20	0.74	3.18	3.83	0.67	3.80
Joint 20:1	2.25	0.74	3.20	3.79	0.68	3.80

3. EXPERIMENTAL SETUP

3.1. Data and Controlled Training

A frozen Ooblect waveform VAE, trained in-house from scratch using the Stable Audio 2.0 architecture, encodes 44.1-kHz audio into 64-dimensional latents at 21.53 frames/s (2048-fold downsampling). Captions and transcripts use the fixed Qwen3.5-9B tokenizer and a 248,320-entry model vocabulary. The generator’s shared text embeddings and tied output head are trainable.

The training mixture contains 26.23M entries, including replay, with a 60/40 Chinese/English split. We use caption-plus-transcript conditioning, random-span inpainting, and text-conditioning dropout of 0.2. Seed-TTS-eval [17] contains 2020 Chinese and 1088 English cases. Controlled comparisons use the same reference audio, transcripts, and assignment of two fixed captions per language.

Table 2 compares generation-only, Joint 10:1, and Joint 20:1 continuations from Joint 10:1 300k for 200k updates. Table 3 starts all variants from Joint 10:1 800k EMA for 100k updates. Both reader variants use $K = 10$, $g_{\text{rec}} = 4$, and $(\lambda_G, \lambda_U, \lambda_R) = (10/12, 1/12, 1/12)$. RSUF-off retains λ_G and λ_U . Table 4 provides a practical reference for the reconstruction budget. The converged model (Table 1) continues the frozen-reader configuration for 300k updates, reaching 1.1M total updates.

3.2. Baseline Evaluation

We re-evaluate six released systems: Qwen2.5-Omni-7B [12], MiMo-Audio-7B-Instruct [13], Step-Audio-2-mini [14], UniAudio 2.0 [4], Ming-Lite-Omni [15], and Ming-UniAudio-16B-A3B [16]. Qwen uses the fixed Chelsie voice and transcript-constrained greedy decoding without reference conditioning. MiMo uses reference-prompt TTS with a voice-matching instruction; Step uses reference-conditioned token-to-wave decoding.

Ming-Lite’s zero-shot Talker uses reference speech, transcript, and speaker features. UniAudio combines `llm_ep2` and Reasoning-Codec for reference-conditioned TTS with a neutral-style instruction

and 10 codec steps. Ming-UniAudio uses native text normalization and reference audio/transcript, with 10 diffusion iterations and CFG=2 for 16-kHz waveform generation.

Baseline generation uses an utterance-specific seed derived from 42. Ming-Lite yields 1086 English outputs after two decoder failures; all other evaluations are complete. One UniAudio case required retrying invalid codec output with another seed. All systems share evaluators but retain their generation interfaces.

Parameter counts exclude standalone codecs/vocoders and the training reader. They include Qwen’s token-to-wave module, Ming-Lite’s Talker, and all MoE experts; tied weights count once.

3.3. Generation and Metrics

For each system and language, MOS averages 160 blinded ratings: 20 listeners each rate eight utterances on a 1–5 quality scale.

Our model uses EMA weights and per-utterance sampling with $v_{\text{cfg}} = (1 + g)v_{\text{cond}} - gv_{\text{uncond}}$, $g = 4$, and zeroed reference in the unconditional branch. We use 48 Euler steps for Chinese and 64 for English, time shift 1, and the same VAE posterior-sampling procedure to encode reference audio as conditioning latents.

We compute Chinese CER with Paraformer [18] and English WER with Whisper large-v3 [19] through *faster-whisper*. SIM uses WavLM-Large features [20] with ECAPA-TDNN [21] (`wavlm_large_finetune.pth`) and cosine similarity to paired reference audio; UTMOS [22] predicts naturalness. Automatic metrics are averaged over utterances. Scores for our checkpoints in Tables 1–3 are further averaged over three inference seeds.

3.4. Auxiliary Protocols

Historical runs (Fig. 2) use learning rate 5×10^{-4} , seed 42, dynamic batches, and 2048 understanding tokens. Generation-only and Joint 10:1 share initialization; Joint 20:1 branches at 300k. The inference pilot (Table 4) uses fixed Joint 800k EMA weights to generate from noise with 4–16 Euler steps, CFG=4, batch size five, preserved punctuation, and historical reference audio.

Table 3. RSUF ablation: 100k updates from Joint 10:1 800k EMA, averaged over three inference seeds. Both reader variants use $K = 10$ and reconstruction guidance $g_{\text{rec}} = 4$. RSUF-off retains shared understanding supervision, removing only $\lambda_R \mathcal{L}_R$.

Training setting	Chinese			English		
	CER (%) ↓	SIM ↑	UTMOS ↑	WER (%) ↓	SIM ↑	UTMOS ↑
RSUF-off	1.88	0.75	3.18	3.39	0.65	3.86
RSUF (unfrozen reader)	1.86	0.77	3.16	3.52	0.67	3.85
RSUF (frozen reader)	1.76	0.77	3.27	3.12	0.67	3.90

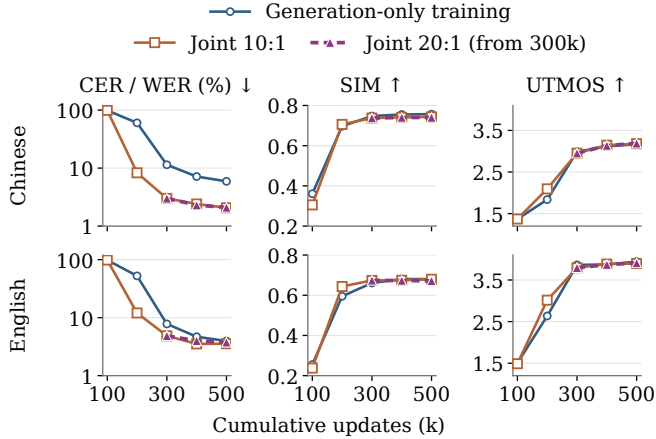


Fig. 2. Historical trajectories (seed 42; log CER/WER). Joint 20:1 branches at 300k with optimizer/data progress reset.

4. RESULTS AND DISCUSSION

Effect of shared understanding supervision. Table 2 tests three continuations from Joint 10:1 300k with another 200k updates each. Relative to generation-only continuation, Joint 10:1 reduces Chinese CER/English WER from 2.52%/4.23% to 2.20%/3.83%. Joint 20:1 reaches 2.25%/3.79%, with UTMOS rising from 3.12/3.76 to 3.20/3.80. Changing the loss-weight ratio has smaller, language-dependent effects. These results support retaining understanding supervision during generation-focused training, with consistent content-accuracy gains across both loss-weight ratios.

RSUF improves on shared understanding supervision. The frozen-reader variant reduces CER/WER from 1.88%/3.39% to 1.76%/3.12% versus RSUF-off (6.4%/8.0% relative; Table 3). It also raises UTMOS from 3.18/3.86 to 3.27/3.90 and SIM from 0.75/0.65 to 0.77/0.67. RSUF evaluates the transcript recoverable from reconstructed speech, adding a content constraint to acoustic reconstruction. Flow matching fits the target velocity at a sampled time, whereas the reader evaluates whether the integrated endpoint preserves the transcript. Differentiating through this endpoint connects local generation updates to an utterance-level content objective. The gains in content accuracy, predicted naturalness, and speaker similarity support this additional supervision beyond shared understanding supervision.

Compared with jointly updating the reader, freezing it yields lower CER/WER and higher UTMOS in both languages, with identical SIM at the reported precision. Both variants use the same reconstruction depth, guidance strength, and loss weights, so the comparison isolates the reader update rule. Keeping the audio-to-text mapping fixed provides a stable target for transcript feedback as the generator changes. These results support the frozen-reader design. Evaluation with separate waveform recognizers further shows that the content gains extend beyond agreement with the training reader.

Table 4. Inference sampling-depth pilot using Joint 800k EMA with CFG=4 (two velocity evaluations per Euler step). * marks the 10-step setting used as a reconstruction-budget reference; bold marks row-wise best values, including ties. Errors (%); ZH/EN: Chinese/English.

Euler steps	4	5	8	10*	12	14	16
ZH CER ↓	3.57	2.65	1.98	1.89	1.89	1.85	1.88
ZH SIM ↑	0.67	0.69	0.73	0.73	0.73	0.73	0.73
ZH UTMOS ↑	2.47	2.95	3.20	3.23	3.24	3.24	3.24
EN WER ↓	3.92	3.58	3.47	3.41	3.29	3.35	3.31
EN SIM ↑	0.60	0.62	0.62	0.63	0.62	0.62	0.62
EN UTMOS ↑	3.49	3.73	3.91	3.92	3.94	3.95	3.94

Converged accuracy and speaker preservation. The converged model reaches 1.69% CER and 3.02% WER and achieves the highest speaker similarity in both languages among the systems in Table 1. It also improves all three automatic metrics over MiMo-Audio and Uni-Audio 2.0 in both languages. These results demonstrate its combination of content accuracy and speaker preservation under the evaluated generation settings. Matched continuations (Tables 2 and 3) isolate the contributions of shared understanding supervision and RSUF.

Learning trajectories. The historical trajectories (Fig. 2) show earlier reductions in transcription errors under joint learning, complementing the matched-continuation results in Table 2. Historical runs track learning across checkpoints; matched continuations isolate the benefit of retaining shared understanding supervision.

Inference sampling depth. Table 4 shows how inference-time generation quality varies with the number of function evaluations (NFE). Increasing Euler steps from 4 to 10 reduces CER/WER from 3.57%/3.92% to 1.89%/3.41%, while SIM and UTMOS approach saturation. Beyond 10 steps, 12–16 steps require 20–60% more evaluations for at most 0.04/0.12 percentage-point further CER/WER reductions, with limited gains elsewhere. The inference-time sweep provides a practical reference for the reconstruction budget. We use $K = 10$ for reconstruction in all RSUF experiments.

5. CONCLUSION

RSUF uses a frozen understanding branch to supervise the content of differentially reconstructed speech. Transcript-loss gradients pass through reconstruction to the generator, with no waveform decoding in the feedback branch or additional inference computation. Matched bilingual continuations support retaining shared understanding supervision in the evaluated model. A separate ablation shows further gains in content accuracy, predicted naturalness, and speaker similarity with RSUF. The reader ablation supports a fixed audio-to-text mapping, and evaluation with separate waveform recognizers shows that content gains extend beyond agreement with the training reader. These findings support combining shared understanding supervision with direct content feedback for generator training.

6. GENERATIVE AI USE DISCLOSURE

Generative AI tools were used only for minor improvements in language and presentation. No AI system was used to generate, modify, or interpret the scientific content of this manuscript.

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